

Mandatory assignment I

1.2.3

$$u(t, x, y) = e^{i(k_x x + k_y y - \omega t)} \quad (*)$$

→ From $u(t, x, y) = \sin(k_x x) \sin(k_y y) \cos(\omega t)$:

$$u_t = -\omega \sin(k_x x) \sin(k_y y) \sin(\omega t)$$

$$u_{tt} = -\omega^2 \sin(k_x x) \sin(k_y y) \cos(\omega t) = -\omega^2 u(t, x, y)$$

$$u_x = k_x \cos(k_x x) \sin(k_y y) \cos(\omega t)$$

$$u_{xx} = -k_x^2 \sin(k_x x) \sin(k_y y) \cos(\omega t) = -k_x^2 u(t, x, y)$$

and similarly,

$$u_{yy} = -k_y^2 u(t, x, y)$$

Going to the wave equation $\frac{\partial^2 u}{\partial t^2} = c^2 \nabla^2 u$:

$$-\omega^2 u(t, x, y) = c^2 (-k_x^2 u(t, x, y) - k_y^2 u(t, x, y)) \Rightarrow$$

$$\Rightarrow \omega^2 = c^2 (k_x^2 + k_y^2) \Rightarrow \omega = c \sqrt{k_x^2 + k_y^2}$$

Now moving to (*):

$$u_t = -i\omega e^{i(k_x x + k_y y - \omega t)}$$

$$u_{tt} = (-i\omega)(-i\omega) e^{i(k_x x + k_y y - \omega t)} = -\omega^2 u(t, x, y)$$

$$u_x = i k_x e^{i(k_x x + k_y y - \omega t)}$$

$$u_{xx} = i k_x i k_x e^{i(k_x x + k_y y - \omega t)} = -k_x^2 u(t, x, y)$$

and similarly,

$$u_{yy} = -k_y^2 u(t, x, y)$$

And now, going to the wave equation:

$$-\omega^2 u(t, x, y) = c^2 (-k_x^2 u(t, x, y) - k_y^2 u(t, x, y)) \Rightarrow$$

$$\Rightarrow \omega^2 = c^2 (k_x^2 + k_y^2) \Rightarrow \omega = c \sqrt{k_x^2 + k_y^2}$$

And so, (*) satisfies the wave equation because the dispersion relation is the same as the stationary wave's.

So (*) is an exact solution of the wave equation.

1.2.4

$$\frac{u_{ij}^{n+1} - 2u_{ij}^n + u_{ij}^{n-1}}{\Delta t^2} = c^2 \left(\frac{u_{i+1,j}^n - 2u_{ij}^n + u_{i-1,j}^n}{h^2} + \frac{u_{i,j+1}^n - 2u_{ij}^n + u_{i,j-1}^n}{h^2} \right) \quad (*)$$

and $u_{ij}^n = e^{i(kh(i+j) - \tilde{\omega}n\Delta t)}$

First,

$$\begin{aligned} u_{i+1,j}^n &= e^{i(kh(i+1+j) - \tilde{\omega}n\Delta t)} = e^{i(kh + kh(i+j) - \tilde{\omega}n\Delta t)} = \\ &= e^{ikh} e^{i(kh(i+j) - \tilde{\omega}n\Delta t)} = e^{ikh} u_{ij}^n \\ u_{i-1,j}^n &= e^{i(kh(i-1+j) - \tilde{\omega}n\Delta t)} = e^{-ikh} u_{ij}^n \end{aligned}$$

Similarly, $u_{i,j+1}^n = e^{ikh} u_{ij}^n$ and $u_{i,j-1}^n = e^{-ikh} u_{ij}^n$

Also,

$$u_{ij}^{n+1} = e^{i(kh(i+j) - \tilde{\omega}(n+1)\Delta t)} = e^{i(kh(i+j) - \tilde{\omega}n\Delta t - \tilde{\omega}\Delta t)} = e^{-i\tilde{\omega}\Delta t} u_{ij}^n$$

and

$$u_{ij}^{n-1} = e^{i(kh(i+j) - \tilde{\omega}(n-1)\Delta t)} = e^{i\tilde{\omega}\Delta t} u_{ij}^n$$

Substituting into (*):

$$(*) \Rightarrow \frac{e^{-i\tilde{\omega}\Delta t} u_{ij}^n - 2u_{ij}^n + e^{i\tilde{\omega}\Delta t} u_{ij}^n}{\Delta t^2} = c^2 \left(\frac{e^{ikh} u_{ij}^n - 2u_{ij}^n + e^{-ikh} u_{ij}^n}{h^2} + \frac{e^{ikh} u_{ij}^n - 2u_{ij}^n + e^{-ikh} u_{ij}^n}{h^2} \right)$$

$$\Rightarrow u_{ij}^n \frac{(e^{-i\tilde{\omega}\Delta t} - 2 + e^{i\tilde{\omega}\Delta t})}{\Delta t^2} = c^2 u_{ij}^n \left(\frac{e^{ikh} - 2 + e^{-ikh}}{h^2} + \frac{e^{ikh} - 2 + e^{-ikh}}{h^2} \right) \Rightarrow$$

$$\Rightarrow \frac{e^{-i\tilde{\omega}\Delta t} - 2 + e^{i\tilde{\omega}\Delta t}}{\Delta t^2} = \frac{2c^2}{h^2} (e^{ikh} - 2 + e^{-ikh}) \quad *$$

$$\Rightarrow 2\cos(\tilde{\omega}\Delta t) - 2 = \frac{2c^2}{h^2} \Delta t^2 (2\cos(kh) - 2) \Rightarrow \cos(\tilde{\omega}\Delta t) - 1 = \frac{2c^2}{h^2} \Delta t^2 (\cos(kh) - 1) \quad **$$

$$\Rightarrow \cos(\tilde{\omega}\Delta t) - 1 = 2C^2 (\cos(kh) - 1) \Rightarrow$$

$$\Rightarrow \cos(\tilde{\omega}\Delta t) - 1 = 2C^2 \cdot (-2\sin^2(\frac{kh}{2})) \Rightarrow$$

$$\Rightarrow \cos(\tilde{\omega}\Delta t) - 1 = -4C^2 \sin^2(\frac{kh}{2}) \Rightarrow -2\sin^2(\frac{\tilde{\omega}\Delta t}{2}) = -4C^2 \sin^2(\frac{kh}{2}) \Rightarrow$$

$$\Rightarrow \sin^2(\frac{\tilde{\omega}\Delta t}{2}) = 2C^2 \sin^2(\frac{kh}{2}) \Rightarrow \sin(\frac{\tilde{\omega}\Delta t}{2}) = \sqrt{2}C \sin(\frac{kh}{2}) \Rightarrow$$

$$\Rightarrow \frac{\tilde{\omega}\Delta t}{2} = \arcsin(\sqrt{2}C \sin(\frac{kh}{2})) \Rightarrow \tilde{\omega} = \frac{2}{\Delta t} \arcsin(\sqrt{2}C \sin(\frac{kh}{2}))$$

For $C = \frac{1}{\sqrt{2}}$: $\tilde{\omega} = \frac{2}{\Delta t} \arcsin(\sin(\frac{kh}{2})) = \frac{2}{\Delta t} \frac{kh}{2} = \frac{kh}{\Delta t}$

We know that $C = \frac{c\Delta t}{h} \Rightarrow \frac{1}{\sqrt{2}} = \frac{c\Delta t}{h} \Rightarrow h = \sqrt{2}c\Delta t \quad (**)$

and also the exact dispersion relation is $\omega = c\sqrt{k_x^2 + k_y^2}$
but since $k_x = k_y = k$:

$$\omega = c\sqrt{k^2 + k^2} \Rightarrow \omega = c\sqrt{2}k \Rightarrow c = \frac{\omega}{\sqrt{2}k}$$

Back to c **(**)** $\Rightarrow h = \sqrt{2} \frac{\omega}{\sqrt{2}k} \Delta t \Rightarrow h = \frac{\omega \Delta t}{k}$

And now, $\tilde{\omega} = \frac{kh}{\Delta t} = \frac{k}{\Delta t} \frac{\omega \Delta t}{k} = \omega \quad \checkmark$

$$\star \cos(z) = \frac{e^{iz} + e^{-iz}}{2}$$

$$\star\star C = \frac{c\Delta t}{h}$$